Fiber-Guided Support Refinement and Type Negation

1 Support padding

A qualifier is represented by a declared finite support and a predicate over stores on exactly that support:

$$\phi = (D_\phi, [\![\phi]\!]), \qquad D_\phi = \mathsf{FV}(\phi).$$

For a finite set X, define support padding $X.\{\phi\}$ by

$$\mathsf{FV}(X.\{\phi\}) = X \cup \mathsf{FV}(\phi), \qquad \sigma \models X.\{\phi\} \iff (\sigma\restriction_{\mathsf{FV}(\phi)}) \models \phi.$$

Padding changes what is observed, not the predicate after projection. In particular, $\{x\}.\{\top\}$ is true on every store over x, but unlike an empty-support top it explicitly observes x.

Expected equations include

$$\varnothing.\{\phi\} \equiv \phi, \qquad X.\{Y.\{\phi\}\} \equiv (X \cup Y).\{\phi\}, \qquad X \subseteq \mathsf{FV}(\phi) \Rightarrow X.\{\phi\} \equiv \phi.$$

Here equivalence includes agreement of declared support, not merely pointwise logical equivalence.

2 The right-facing fiber operator

For a finite set X, the formula $X \triangleright P$ is interpreted by fixing the values of X and checking the substituted formula in each corresponding fiber:

$$r \models X \triangleright P \quad \iff \quad \forall \sigma_X \in r\restriction_X.\ \mathsf{Fiber}(r, X, \sigma_X) \models \sigma_X(P). \tag{1}$$

This is the paper-level form of the artifact's `FFibVars` connective.

At formula level, write the right-facing fiber composition as

$$P \blacktriangleright Q \ \triangleq\ P \wedge \mathsf{FV}(P) \triangleright Q. \tag{2}$$

The order matters: Q may depend on choices exposed by P.

3 Relationship with separating conjunction

Under the artifact's Cartesian resource product, separating conjunction entails both orders of fiber composition:

$$P * Q \models P \blacktriangleright Q, \qquad P * Q \models Q \blacktriangleright P. \tag{3}$$

The product contains every compatible combination of a P-store and a Q-store. Fixing either support therefore preserves all choices supplied by the other operand.

There is an exact fragment in which both implications reverse. Let

$$P \triangleq x : [b_1 \mid \phi], \qquad Q \triangleq y : [b_2 \mid \psi], \qquad x \neq y,$$

where both formulas are atomic Under bindings, their supports are disjoint, and both typed qualifiers are inhabited. Define their canonical covered value sets A and B. Under requires every point in A and B to be covered. Hence either directional fiber composition contains the complete rectangle $A \times B$, exactly the canonical product required by $P * Q$:

$$A \triangleq \{v \mid v : b_1 \wedge \phi(v)\}, \qquad B \triangleq \{w \mid w : b_2 \wedge \psi(w)\}, \qquad A \times B \subseteq r\restriction_{\{x,y\}}.$$

$$P * Q \iff P \blacktriangleright Q \iff Q \blacktriangleright P. \tag{4}$$

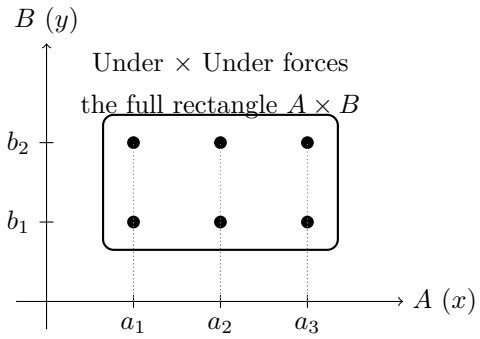

图 1: For disjoint inhabited atomic Under formulas, every A-fiber covers all of B, so fiber composition and separating product coincide.

For general formulas, the reverse implications fail. The simplest failure uses an Over atom. Let

$$P \triangleq x : \{\mathsf{Bool} \mid \top\}, \qquad Q \triangleq y : [\mathsf{Bool} \mid \top],$$

and take the diagonal capability

$$r_{\mathrm{diag}} = \{(x = \mathsf{false}, y = \mathsf{false}), (x = \mathsf{true}, y = \mathsf{true})\}.$$

Then $r_{\mathrm{diag}} \models Q \blacktriangleright P$: Q covers both values of y, and each fixed-y fiber contains one safe Boolean x, which is enough for the Over atom P. However,

$$r_{\mathrm{diag}} \not\models P * Q.$$

The Under resource for Q must contain both y values, and its Cartesian product with any nonempty Over witness for P contains an off-diagonal point. Thus $*$ is strictly stronger in general; Under is the exact fragment because it forces the same complete canonical set in every fiber.

4 Dependent type negation

No new wand is needed: ordinary dependent arrows can repeatedly expose the increasing fiber context. Consider a function whose parameters and result are all Over base types:

$$x : \{\mathsf{Nat} \mid \top\} \to y : \{\mathsf{Nat} \mid x < \nu\} \to \{\mathsf{Bool} \mid \psi(x, y, \nu)\}. \tag{5}$$

A traditional negation changes each Over parameter into a disjoint Under parameter and complements the result qualifier:

$$x : [\mathsf{Nat} \mid \top] \mathrel{-\!\!*} y : [\mathsf{Nat} \mid x < \nu] \mathrel{-\!\!*} [\mathsf{Bool} \mid \neg\psi(x, y, \nu)]. \tag{6}$$

The second separating premise is inappropriate: its resource must be disjoint from x, while its qualifier simultaneously reads x.

Fiber refinement instead keeps ordinary arrows and accumulates all previously visible variables in both later parameters and the final Under result:

$$x : [\mathsf{Nat} \mid \top] \to y : [\mathsf{Nat} \mid \{x\}.\{x < \nu\}] \to [\mathsf{Bool} \mid \{x, y\}.\{\neg\psi(x, y, \nu)\}]. \tag{7}$$

At formula level the parameter context is

$$(x : [\mathsf{Nat} \mid \top]) \blacktriangleright (y : [\mathsf{Nat} \mid x < \nu]).$$

For a further parameter z depending on x and y, the translation uses

$$z : [b \mid \{x, y\}.\{\psi(x, y, \nu)\}],$$

or, equivalently, one more right-facing composition. Negation therefore changes every Over parameter into Under coverage and the final Over result into the Under complement. Dependencies remain expressible because each successive type grows its fiber observation set instead of imposing disjointness.

5 Proof obligations

A formal development should establish support-padding normalization, the two entailments in (3), and the atomic-Under characterization (4). It should then formalize the Over diagonal counterexample and prove that the fiber-accumulating translation (7) correctly implements type negation for dependent parameters.